BAOS Process CHECKER

Version **v2.0.1** · Built **2026-11-12** · Author **Kwag D.S**

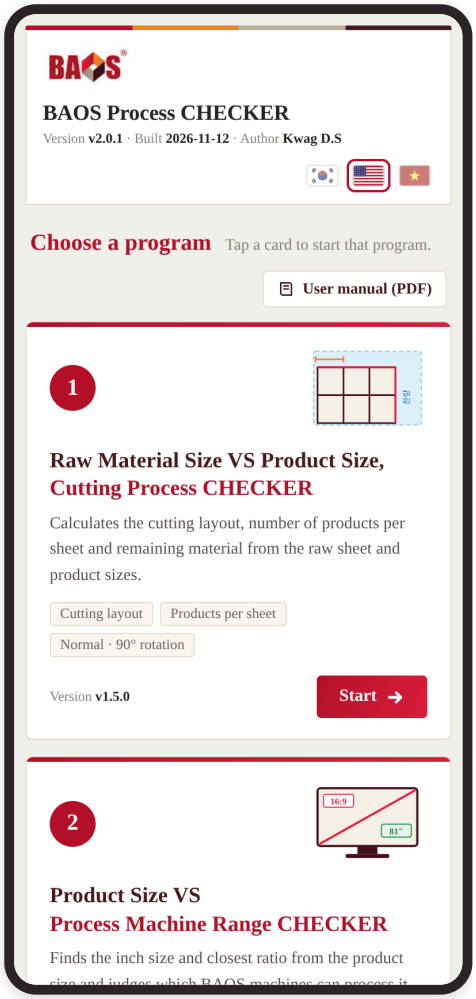
User Manual

Mobile app · using the link received in KakaoTalk on iPhone and Android

Web address

Point your phone camera at the QR code to open it

`https://orrimaster.github.io/baos-Assistant/`



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1 What does this program do?

BAOS Process CHECKER brings two calculations often needed on the production floor together in one place. Choose the program you need on the main menu.

1 Raw Material Size VS Product Size, Cutting Process CHECKER

Enter the raw sheet size and the product size to see **how many products can be cut from one sheet**, which orientation is better and how much material is left, together with a **cutting layout drawing**.

2 Product Size VS Process Machine Range CHECKER

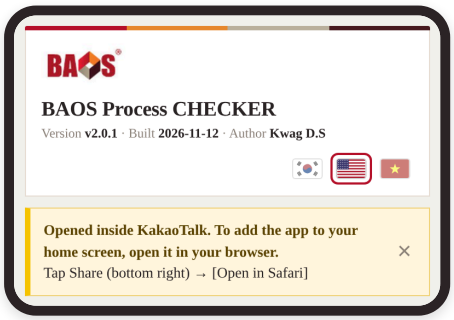
Enter the product size (length and width) to see the **diagonal and inch size**, the **closest screen ratio** and the **list of BAOS machines that can process it**.

- Use the **flag buttons** to switch between 한국어 / English / Tiếng Việt. The choice applies to the menu and both programs.
- Press the **house button** at the top right of each program to return to the main menu.
- Results update as soon as you type. There is no calculate button.
- Values you enter are calculated only on your device and are never sent outside.
- Machine data source: **20261112 VS DB FORM Ver4** (26 machines)

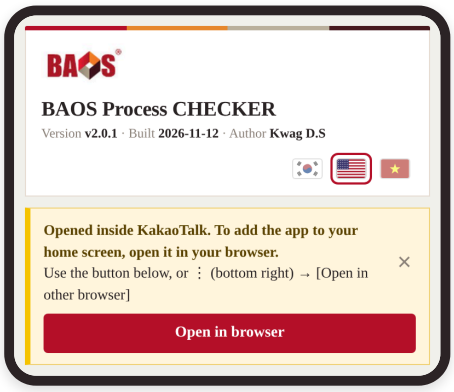
2 Opening the link from KakaoTalk

Tap the link your administrator sent in KakaoTalk and the checker opens. It is the same on iPhone and Android.

1. In the KakaoTalk chat room, tap the **blue link** (`orrimaster.github.io/baos-Assistant`).
2. The main menu opens inside KakaoTalk (in-app browser). **You can already use it as it is.**
3. A **yellow notice bar** appears at the top. To add the app to your home screen, reopen it in your browser as the notice says. (Chapter 3)



iPhone · notice shown inside KakaoTalk



Android · notice shown inside KakaoTalk

Why add it to the home screen? You will not need to search KakaoTalk for the link every time. One tap on the icon opens it full-screen without an address bar, even without internet.

TIP · Keep the link Long-press the link in the chat, tap [Copy] (복사) and paste it into **"Chat with myself"** (나와의 채팅) so you can always find it. Close the yellow bar with × if you do not need it.

3 Adding the app to your home screen

Follow the side that matches your phone. You only need to do this once.

iPhone (Safari)

1. In KakaoTalk, tap the **Share button at the bottom right** and choose **[Open in Safari]** (Safari로 열기).
2. When the main menu opens in Safari, tap the **Share button** (square with an up arrow) at the bottom.
   On iOS 26 or later, if you do not see it, tap ⋯ at the bottom → [Share].
3. Scroll the menu up and tap **[Add to Home Screen]**.
4. Check that the name is **“BAOS CHECKER”** and tap **[Add]** at the top right.
5. Tap the new **BAOS icon** on your home screen to open it like an app.

Android (Chrome)

1. In KakaoTalk, tap **[Open in other browser]** (다른 브라우저로 열기) on the yellow bar.
   Or tap ⋮ at the bottom right → [Open in other browser]
2. When the main menu opens in Chrome, tap ⋮ **at the top right**.
3. Tap **[Add to Home screen]** or **[Install app]**.
   In the latest Chrome it may be called [Install and create shortcut].
4. Tap **[Install]** (or [Add]).
5. Tap the new **BAOS icon** on your home screen (or app list) to open it like an app.

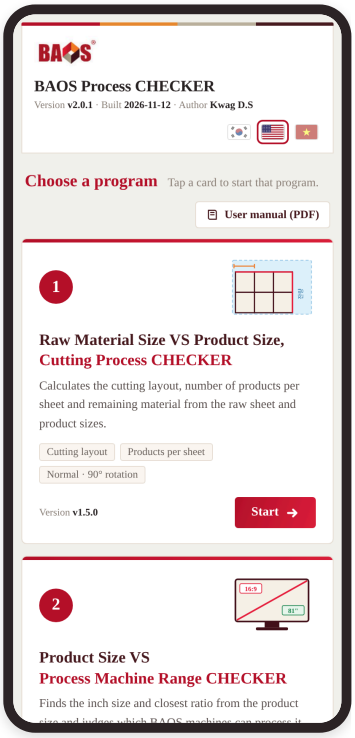


This icon appears on your home screen with the name **“BAOS CHECKER”**.

If it opened in Samsung Internet (Galaxy) Tap ≡ at the bottom → [Add page to] → [Home screen]. If an install icon appears in the address bar, you can tap that instead. Menu names may differ slightly by phone and browser version.

TIP · Open with the QR code Point your phone’s **Camera app** at the QR code on the cover and tap the link that appears. It opens directly in Safari or Chrome, so skip step 1 above and start from step 2.

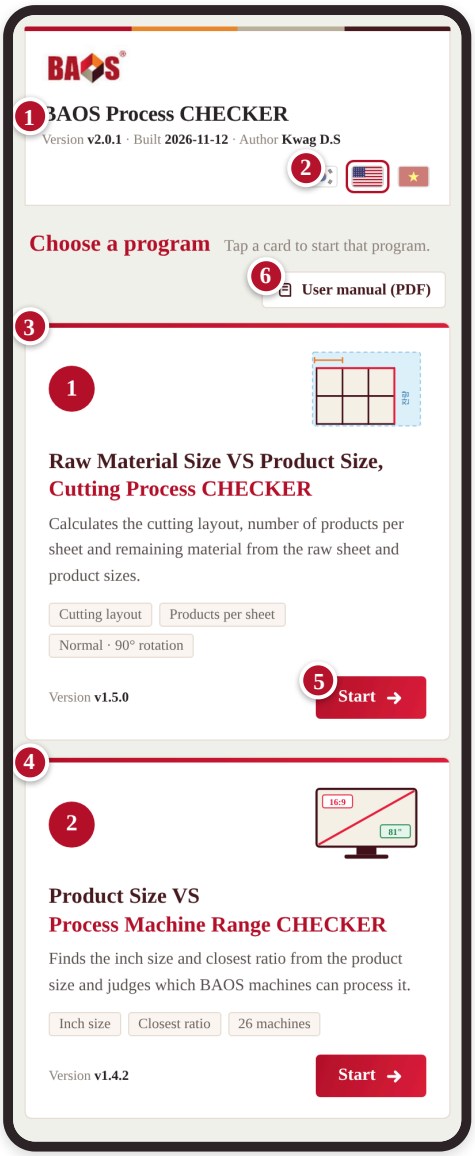
On a PC Type the web address in the Edge or Chrome address bar, then click the install icon at the right of the address bar → [Install] to install it as an app.



Opened from the home screen icon

4 Main menu

This is the first screen when the program opens. Tap the card of the program you want to use.



No.		
1	**Title**	Program name (BAOS Process CHECKER), version, build date, author
2	**Flags (language)**	한국어 / English / Tiếng Việt. The red frame shows the current language
3	**Card ①**	Starts Raw Material Size VS Product Size, Cutting Process CHECKER

No.		
4	**Card ②**	Starts Product Size VS Process Machine Range CHECKER
5	**Start button**	Tapping anywhere on the card starts the same program
6	**User manual**	Opens this manual (PDF) in the current language

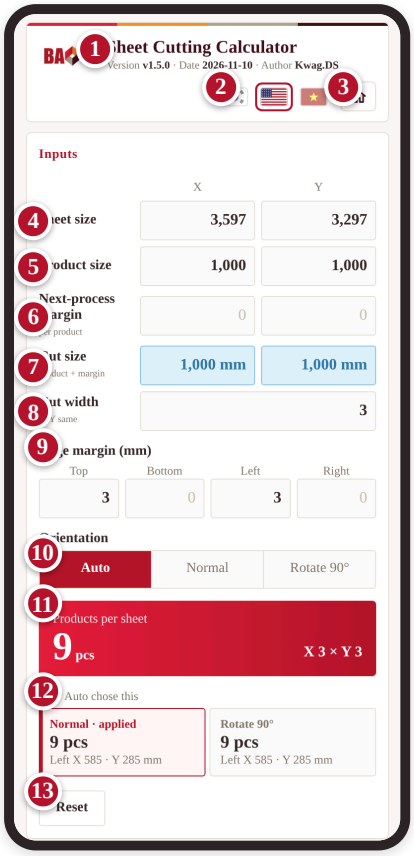
Controls on every screen

- **Flags**: change the language. It applies to the menu and both programs and is remembered next time.
- **House button** (top right of each program, next to the flags): returns to the main menu.
- You can also go back with the phone's **Back** button (Android) or by swiping right from the left edge (iPhone).

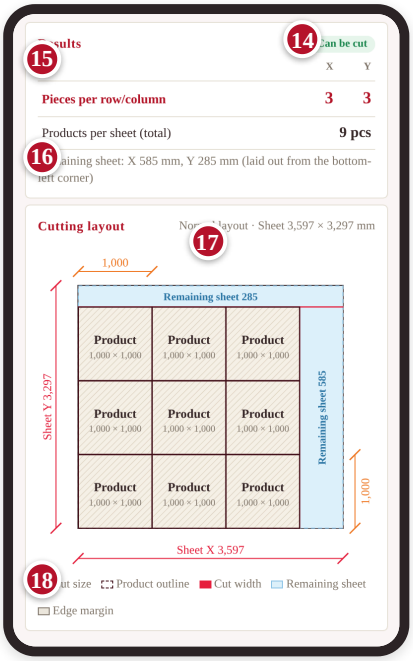
5 Program ① Screen layout

On a phone the screen runs from top to bottom: inputs → products per sheet → results → layout drawing.

① Top: inputs



② Bottom: results · drawing



No.		
1	**Title**	Program name (Sheet Cutting Calculator), version, date, author
2	**Flags (language)**	Switch 한국어 / English / Tiếng Việt
3	**House button**	Back to the main menu
4	**Sheet size**	X (length) · Y (width) of the raw sheet, mm
5	**Product size**	X · Y of the finished product, mm
6	**Next-process margin**	Allowance per product removed in the next process (X · Y). Blank = 0
7	**Cut size**	Product + margin. Calculated automatically (light-blue box)
8	**Cut width**	Width lost where the saw passes. Same for X · Y. Blank = 0
9	**Edge margin**	Width discarded at the top · bottom · left · right edges of the sheet

No.		
10	**Orientation**	Auto / Normal / Rotate 90°
11	**Products per sheet**	Products from one sheet (X count × Y count)
12	**Why Auto chose this**	Comparison of quantity and remaining sheet for Normal and Rotate 90°
13	**Reset · Print**	Return to default values / print
14	**Status**	Can be cut · Caution · Sheet too small · Waiting for input
15	**Pieces per row/column**	Counts in X and Y and the total
16	**Remaining sheet note**	Size of leftover material, or how much is missing
17	**Cutting layout**	Layout drawing (filled from the bottom-left corner)
18	**Legend**	Meaning of the drawing colours

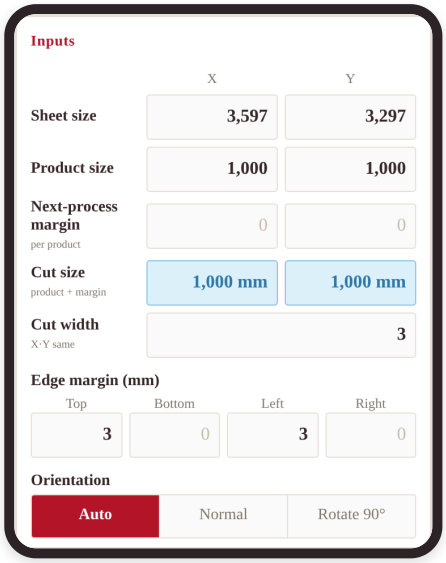
6 Program ① Step-by-step use

Example: cutting products of **1,000 × 1,000 mm** from a sheet of **3,597 × 3,297 mm** (the program's default values).

STEP 1 Enter the sheet size

1. Type **3597** in the X box and **3297** in the Y box of **Sheet size**.

- X is the length, Y the width, in mm.
- When you leave the box, a thousands separator is added (**3,597**).
- Press **Enter** to move to the next box.



STEP 2 Enter the product size and margin

1. Type **1000** / **1000** in **Product size**.
2. Leave **Next-process margin** blank if there is none (counts as 0).

- The **Cut size** (light-blue boxes) shows product + margin automatically — 1,000 mm × 1,000 mm here.
- The margin is a value **per product**. Example: if 2.5 mm is removed on every side in the next process, enter 5 for X and Y.

STEP 3 Enter the cut width and edge margin

1. Type **3** in **Cut width** (saw thickness).
2. In **Edge margin**, type **3** for Top and **3** for Left; leave Bottom and Right blank.

- The edge margin cuts away bent or rough edges of the sheet.
- **At least 2 of top, bottom, left and right need 3 mm or more** to set a square reference. Otherwise "Caution" appears.

STEP 4 Choose the orientation

- Normally leave it on **Auto**. Both Normal and Rotate 90° are calculated and the better one is chosen.
- To fix the direction, choose **Normal** or **Rotate 90°**. Rotate 90° swaps the product's X and Y.

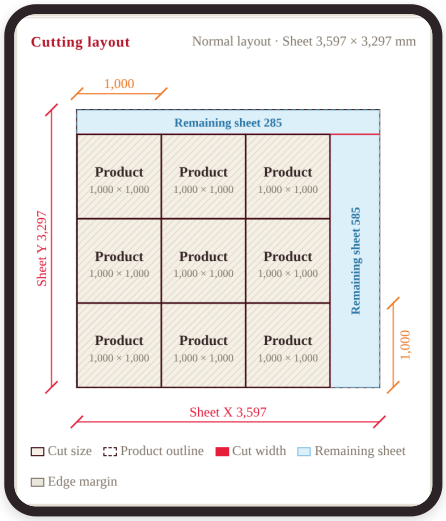
STEP 5 Check the results

- Red box **Products per sheet**: **9** (X 3 × Y 3)
- The status at the top right is green **“Can be cut”**.
- The note shows **Remaining sheet X 585 mm, Y 285 mm**.



STEP 6 Check the cutting layout

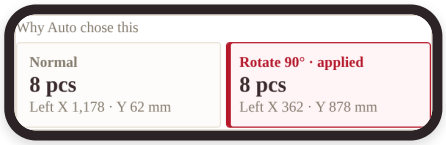
- Products are filled from the **bottom-left corner** of the sheet.
- Hatched cells are the products to cut, red lines the cut width, light blue the remaining sheet and grey edges the edge margin.
- The orange dimension shows the cut size; the red dimensions show the sheet size.



STEP 7 (Example 2) Automatic orientation

Change the product to **1,200 × 800 mm** with a margin of **5 / 5 mm**; the cut size becomes 1,205 × 805 mm.

- Normal: **8**, remaining X 1,178 · Y 62 mm
- Rotate 90°: **8**, remaining X 362 · Y 878 mm
- The quantities are equal, so **Rotate 90°, with more remaining material,** is chosen automatically (marked “applied”).
- Tap a comparison card to switch to that orientation.



7 Program ① Reading the results · how it is calculated

Calculation (done separately for X and Y)

Item	Calculation	Example (X direction)
Cut size	Product + next-process margin	1,000 + 0 = **1,000**
Usable length	Sheet − edge margins (both sides)	3,597 − 3 − 0 = **3,594**
Count	Whole part of (usable length + cut width) ÷ (cut size + cut width)	(3,594 + 3) ÷ (1,000 + 3) = 3.58 → **3**
Leftover length	Usable length − count × cut size − (count − 1) × cut width	3,594 − 3,000 − 6 = **588**
Remaining sheet	Leftover length − cut width (one cut to separate it)	588 − 3 = **585 mm**
Products per sheet	X count × Y count	3 × 3 = **9**

How Auto chooses the orientation

1. First, the layout that **fits** in the sheet.
2. If both fit, the one with **more products**.
3. If equal, the one with **more remaining material** (the smaller of X · Y).
4. If still equal, **Normal**.

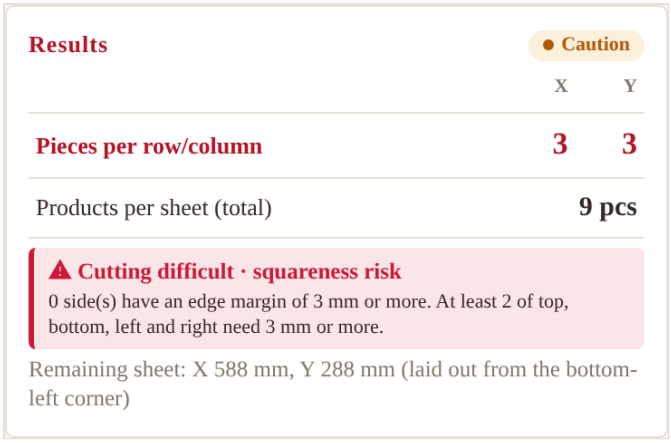
The heading above the comparison cards reads **Why Auto chose this** in Auto mode and **Layout comparison** when you choose the orientation yourself.

Status

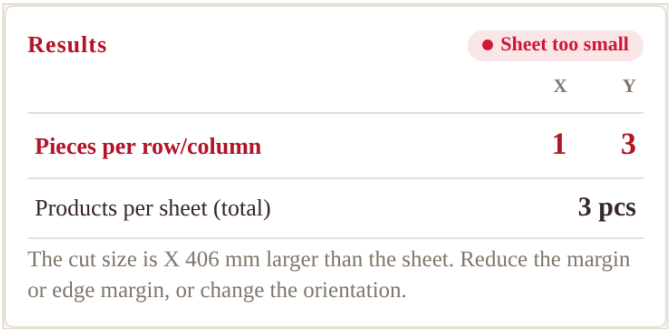
Shown	Meaning	What to do
Can be cut	Everything fits in the sheet and the edge margin is sufficient.	Proceed.
Caution	Cutting is possible, but fewer than 2 sides have an edge margin of 3 mm or more.	Increase the edge margin (see below).
Sheet too small	The cut size is larger than the sheet. The note shows how much is missing.	Reduce margin / edge margin or change the orientation.
Waiting for input	One of sheet X · Y or product X · Y is blank.	Fill in all four boxes.

Edge margin warning · sheet too small examples

- **Cutting difficult · squareness risk**: shown when **fewer than 2 sides** of top, bottom, left and right have an edge margin of **3 mm or more**. The left picture has all edge margins at 0.
- **Sheet too small**: the right picture uses a product of 4,000 × 1,000 mm. The note shows the shortfall, e.g. “406 mm larger”. The count then shows the minimum (1) for reference only.



Edge margin warning (all edges 0)



Sheet too small (product 4,000 × 1,000)

Reading the cutting layout

Hatched cell	Cut size (one product). With 200 cells or fewer, “Product” and its size are written
Dashed rectangle	Product outline (inside the cut size when there is a margin)
Red band	Cut width (where the saw passes)
Light blue	Remaining sheet, with its size
Grey edge	Edge margin (discarded)
Red dimension	Sheet X · Y
Orange dimension	Cut size (one product)
Red-framed cells	Sheet too small

With more than 3,000 cells only the first 3,000 are drawn. The numbers are still correct.

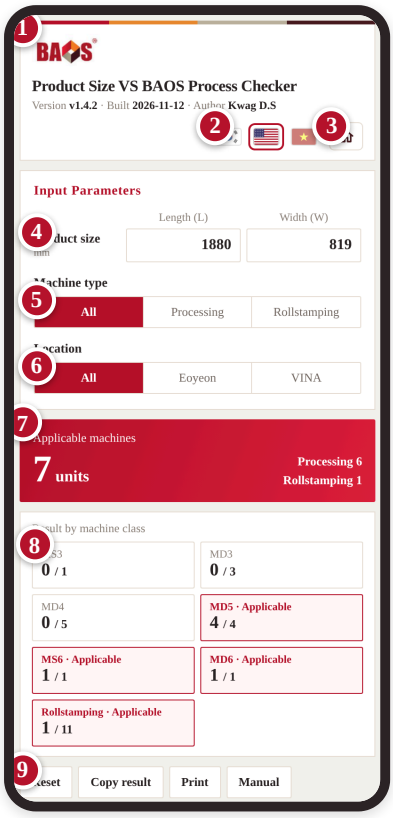
Input tips

- Only numbers can be typed. Thousands separators are added when you leave the box.
- Clicking a box selects the existing number, so you can type a new one straight away.
- Press **Enter** (on a phone, [Next] on the keypad) to move to the next box.
- A **blank** margin, cut width or edge margin counts as **0**.

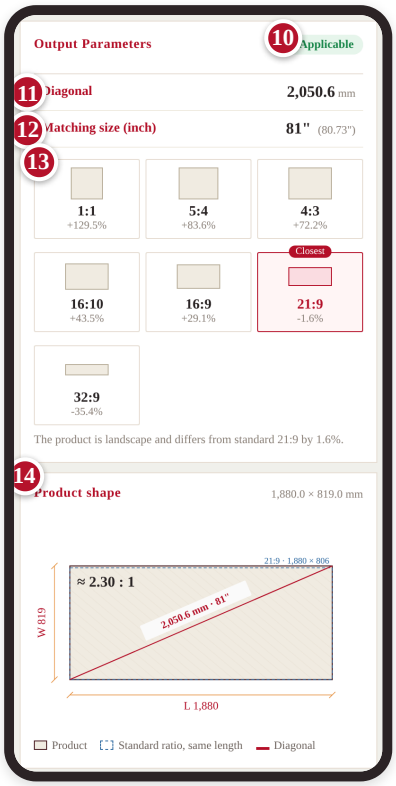
8 Program ② Screen layout

On a phone the screen continues vertically. Scroll down from top to bottom. (The picture below splits one screen into three parts.)

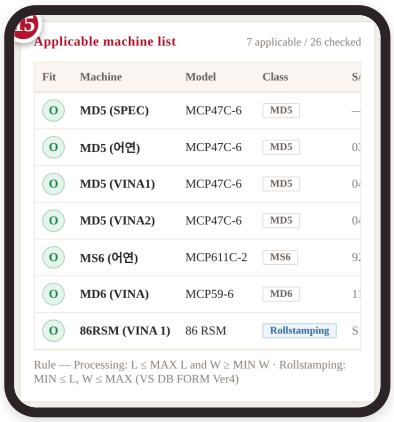
① Top: input



② Middle: results



③ Bottom: machine list



No.		
1	**Title**	Program name, version, build date, author
2	**Flags (language)**	Switch 한국어 / English / Tiếng Việt
3	**House button**	Back to the main menu
4	**Product size**	Boxes for length (L) and width (W) in mm
5	**Machine type**	All / Processing / Rollstamping
6	**Location**	All / Eoyeon / VINA
7	**Applicable machines**	Number of machines that can process the product
8	**Result by machine class**	“Applicable / total” for each class

No.		
9	**Buttons**	Reset · Copy result · Print · Manual
10	**Status badge**	Green “Applicable” / red “Not applicable”
11	**Diagonal**	Length of the product diagonal (mm)
12	**Matching size (inch)**	Rounded inch and actual inch (2 decimals)
13	**Standard ratio cards**	7 standard ratios with difference (%), “Closest” mark
14	**Product shape**	Drawing of the entered product
15	**Applicable machine list**	Only machines judged O are listed

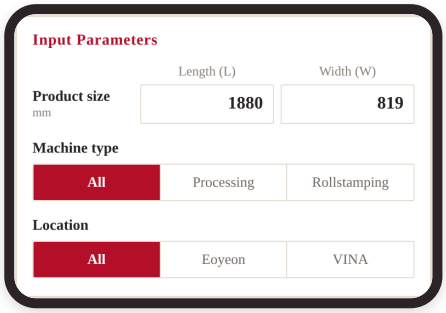
9 Program ② Step-by-step use

Example: checking a product of length **1880 mm** and width **819 mm**.

STEP 1 Enter the product size

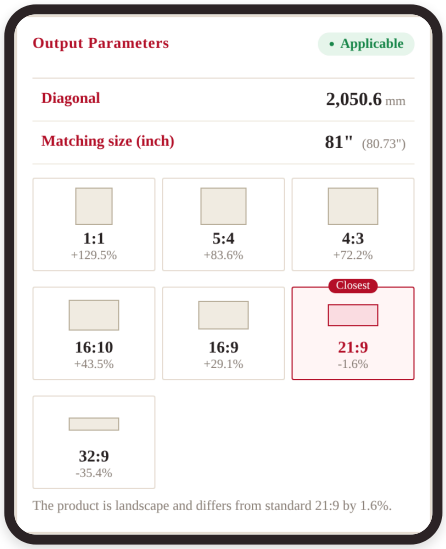
1. Tap the **Length (L)** box; the number keypad appears. Type **1880**.
2. Tap the **Width (W)** box and type **819**.
3. Tap an empty area to close the keypad.

- Results update as soon as you type. There is no calculate button.
- Tapping a box selects the existing number, so you can type a new one without deleting.
- **[Reset]** clears the boxes so you can type in the length box right away.
- **Do not swap length (L) and width (W)** from the drawing — the judgment changes.



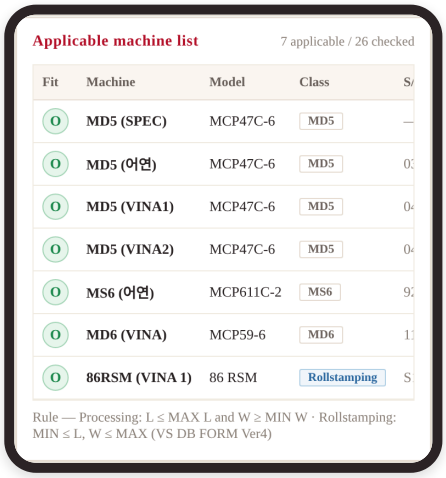
STEP 2 Scroll down to see the results

- **Diagonal**: 2,050.6 mm
- **Matching size**: 81" (actual 80.73")
- A green **“Applicable”** badge at the top right means at least one machine can process the product.
- Among the ratio cards, the one with a red border and **“Closest”** is the nearest standard ratio — **21:9 (-1.6%)** in the example.



STEP 3 Check the applicable machines

- The red box shows the **number of applicable machines** — **7** in the example.
- The **Applicable machine list** at the bottom shows machine name, model, S/N and range.
- If the list is wider than the screen, **swipe left and right** to see all columns.



STEP 4 Share the result in KakaoTalk



1. Tap **[Copy result]**. “Result copied to clipboard” appears briefly.
2. Open the KakaoTalk chat room you want to send to.
3. **Long-press the message box and tap [Paste]** (붙여넣기), then send.

What is sent when you paste

```
[BAOS] Product Size VS BAOS Process Checker
Product size: 1880 × 819 mm
Closest: 21:9 (-1.6%)
Diagonal: 2050.6 mm · Matching size (inch): 81" (80.73")
Applicable machines: 7
- MD5 (SPEC) [MD5]
- MD5 (어연) (#0344) [MD5]
  …
```

10 Program ② Reading the results

Diagonal

The length of the product diagonal, calculated as $\sqrt{(length^2 + width^2)}$ and shown with 1 decimal.

Matching size (inch)

The diagonal (mm) divided by 25.4. The **rounded whole number** is shown large, with the **actual value (2 decimals)** in brackets. Values of .5 and above are rounded up.

Product size (mm)	Diagonal (mm)	Actual inch	Shown as
1,218 × 685	1,397.4	55.02"	**55"** (55.02")
1,880 × 819	2,050.6	80.73"	**81"** (80.73")
700 × 420	816.3	32.14"	**32"** (32.14")
1,650 × 930	1,894.0	74.57"	**75"** (74.57")

Standard ratio cards

- The product ratio (long side ÷ short side) is compared with seven ratios: 1:1, 5:4, 4:3, 16:10, 16:9, 21:9, 32:9.
- The % under each card is the difference from that ratio. + means the product is longer/narrower than the standard, − means shorter/wider.
- The card closest to 0% gets a red border and the **“Closest”** tag. Within ±0.5% it shows “≈ 0%”.
- Portrait products (width longer than length) are also compared by their long side.

Product shape drawing

- **Brown solid rectangle**: the product size you entered
- **Blue dashed rectangle**: the size at the “Closest” ratio with the same length (e.g. 21:9 → 1,880 × 806)
- **Red line**: the diagonal with its length and inch size

Applicable machines · result by class

- The large number in the red box is how many machines can process the product under the current filters. Processing / Rollstamping counts are shown on the right.
- **“4 / 4”** on a class card means “4 applicable / 4 machines in that class”.
- Red-bordered cards are classes with applicable machines; grey cards have none.
- When 0 machines apply, the box turns grey and the badge turns red “Not applicable”.

Columns of the applicable machine list

Column	Meaning
Fit	O = can be processed (only O machines are listed)
Machine	Machine name and location (e.g. MD5 (어연) = Eoyeon)
Model	Machine model (e.g. MCP47C-6)
Class	MS3, MD3, MD4, MD5, MS6, MD6 = Processing / Rollstamping
S/N	Machine serial number (“—” if none)
MIN L×W	Minimum length × minimum width that can be processed (mm)
MAX L×W	Maximum length × maximum width that can be processed (mm)
Reason	“In range” = the product is inside the range
Top-right count	“7 applicable / 26 checked” = applicable machines among those checked

Judgment rules

Judgment uses the MIN/MAX values of each machine in the Excel file **VS DB FORM Ver4**. The conditions differ by machine type.

Machine type	Condition for O (applicable)
Processing MS3 · MD3 · MD4 · MD5 · MS6 · MD6	Length ≤ **MAX L** and Width ≥ **MIN W**
Rollstamping 32 · 55 · 65 · 75 · 86 RSM	**MIN L** ≤ Length ≤ **MAX L** and **MIN W** ≤ Width ≤ **MAX W**

Examples (product 1880 × 819)

Machine	Range	Check	Result
MD5 (어연)	MAX L 2,050 · MIN W 500	1,880 ≤ 2,050 ✓ · 819 ≥ 500 ✓	O
MD4 (SPEC)	MAX L 1,300 · MIN W 400	1,880 > 1,300 × (length too long)	X
86RSM (VINA 1)	L 947–1,944 · W 541–1,121	Length ✓ · Width ✓	O
65RSM (어연1)	L 948–1,450 · W 520–830	1,880 > 1,450 × (length too long)	X

> ! The result is a reference based on the machine specification table. Always confirm with the process engineer before actual production.

Using the filters

When you change a filter, the machine count, class results and machine list all follow that condition.

Filter	Option	Machines shown
Machine type	All	All machines
	Processing	MS3 · MD3 · MD4 · MD5 · MS6 · MD6
	Rollstamping	32 · 55 · 65 · 75 · 86 RSM
Location	All	All machines (including SPEC = standard specification)
	Eoyeon	Machines with “어연” (Eoyeon) in the name
	VINA	Machines with “VINA” in the name

Both filters apply together. Example: type “Rollstamping” + location “VINA” → only RSM machines in VINA. If you see fewer machines than expected, first check that the filters are set to “All”.

11 Buttons

Screen	Button	Function
Main menu	**Card · Start**	Starts that program.
Main menu	**User manual**	Opens the manual (PDF) in the current language.
All	**Flag**	Changes the language (all screens, remembered).
All	**House**	Returns to the main menu.
Cutting	**Reset**	Restores defaults (sheet 3,597 × 3,297, product 1,000 × 1,000, cut width 3, edge margin top · left 3, Auto) and puts the cursor in sheet X.
Cutting	**Print**	Prints the current screen. Hidden on phone screens.
Cutting	**Comparison card**	Switches to that orientation (Normal / Rotate 90°).
Machine range	**Reset**	Clears the size boxes so you can type the length right away. Filters return to “All”.
Machine range	**Copy result**	Copies the result as text for a messenger, e-mail or KakaoTalk.
Machine range	**Print**	Prints the screen or saves it as a PDF.
Machine range	**Manual**	Opens the manual (PDF) in the current language.

12 Troubleshooting (FAQ)

Problem	What to do
The link in KakaoTalk does not open	Check that Wi-Fi or mobile data is on. If it still fails, long-press the link, tap [Copy] and paste it into the Safari or Chrome address bar.
I cannot find [Add to Home Screen]	The page is open inside KakaoTalk. First open it in Safari (iPhone) or Chrome (Android), then look again. (Chapter 3)
[Open in other browser] does nothing	Choose it again from the ⋮ menu at the bottom right, or copy the link and paste it into the Chrome address bar.
The yellow notice bar keeps appearing	It appears only inside KakaoTalk. Close it with ×. It does not appear when you open the app from the home screen icon.
I see an old version	Close the app completely and open it again (swipe it up in the recent apps list). Check the version in the title.
I want to go back to the main menu	Press the **house button** next to the flags at the top right of each program.
Cutting: numbers show “-” and “Waiting for input”	All four boxes — sheet X · Y and product X · Y — must be filled in.
Cutting: “Sheet too small” appears	The cut size is larger than the sheet. Reduce the margin or edge margin, or change the orientation.

Cutting: “Caution” and a squareness warning appear	Fewer than 2 sides have an edge margin of 3 mm or more. Set at least 2 of top, bottom, left and right to 3 mm or more.
Cutting: fewer products than expected	Cut width, edge margins and the next-process margin are all included. Check them and keep the orientation on “Auto”.
Cutting: only part of the drawing appears	With more than 3,000 cells only the first 3,000 are drawn. The numbers are correct.
The number keypad does not appear	Tap exactly inside the length or width box.
“Enter length and width” appears and numbers show “–”	This is normal right after [Reset]. Enter numbers in both boxes.
0 applicable machines	The size is outside every machine’s range. Check that length and width are not swapped and that the filters are set to “All”.
Text is too small	Spread two fingers to zoom in, or turn your phone sideways.
Copy result does not work	Tap the button once more, or take a screenshot and send it.
Machine specifications have changed	Operators cannot edit them. Tell the administrator or the author (Kwag D.S).

13 Program ② Appendix · Machine specification list

Source: 20261112 VS DB FORM Ver4 · machine specification sheet (26 machines) · unit mm · 어연 = Eoyeon

Model	Machine	Class	S/N	MIN L × W	MAX L × W
MCP34-2	**MS3 (Spec)**	MS3	—	425 × 325	1,360 × 760
MCP35C-6	**MD3 (SPEC)**	MD3	—	260 × 260	1,400 × 760
MCP35C-6	**MD3 (어연)**	MD3	0565-1	260 × 260	1,400 × 760
MCP35C-6	**MD3 (VINA)**	MD3	0565-2	260 × 260	1,400 × 760
MCP34-4	**MD4 (SPEC)**	MD4	—	716 × 400	1,300 × 730
MCP34-4	**MD4 (어연 55)**	MD4	8859-3	716 × 398	1,380 × 730
MCP34-4	**MD4 (어연 70)**	MD4	8849-6	716 × 580	1,580 × 730
MCP34-4	**MD4 (어연 75)**	MD4	8859-1	716 × 747	1,680 × 730
MCP34-4	**MD4 (VINA 55)**	MD4	8816	716 × 400	1,300 × 730
MCP47C-6	**MD5 (SPEC)**	MD5	—	900 × 500	2,050 × 1,210
MCP47C-6	**MD5 (어연)**	MD5	0344	900 × 500	2,050 × 1,210
MCP47C-6	**MD5 (VINA1)**	MD5	0469-1	900 × 500	2,050 × 1,210
MCP47C-6	**MD5 (VINA2)**	MD5	0469-2	900 × 500	2,050 × 1,210
MCP611C-2	**MS6 (어연)**	MS6	9222	716 × 414	3,330 × 1,870
MCP59-6	**MD6 (VINA)**	MD6	1176	900 × 600	2,600 × 1,500
32 RSM	**32RSM (어연 D)**	Rollstamping	S2007300	352 × 202	730 × 450
32 RSM	**32RSM (VINA 1)**	Rollstamping	S2005400	352 × 202	730 × 450
32 RSM	**32RSM (VINA 2)**	Rollstamping	S2007400	352 × 202	730 × 450
55 RSM	**55RSM (SPEC)**	Rollstamping	S16015	685 × 380	1,250 × 730
55 RSM	**55RSM (어연 5)**	Rollstamping	S16015	500 × 300	1,250 × 730
65 RSM	**65RSM (어연1)**	Rollstamping	D15010	948 × 520	1,450 × 830
65 RSM	**65RSM (VINA 2)**	Rollstamping	S15045	948 × 520	1,450 × 830
65 RSM	**65RSM (VINA 4)**	Rollstamping	S17009	948 × 340	1,450 × 830
75 RSM	**75RSM (어연4)**	Rollstamping	S16014	894 × 508	1,665 × 945
75 RSM	**75RSM (VINA 2)**	Rollstamping	S16005	894 × 508	1,665 × 945
86 RSM	**86RSM (VINA 1)**	Rollstamping	S1809700	947 × 541	1,944 × 1,121

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